MOTOR HEALTH MONITORING SYSTEM

Using Fuzzy Logic & Genetic Programming • Industrial Predictive Maintenance Engine

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1. PROBLEM

1.1 Industrial Context & Prime Mover Reliability

In industrial manufacturing plants, processing facilities, power generation stations, water distribution networks, and HVAC systems, electric motors consume over 70% of total industrial electrical energy. Induction motors operate continuously under arduous mechanical, thermal, and electrical stressors. Statistical empirical studies by the IEEE Industrial and Commercial Power Systems (IC&CPS;) committee indicate that motor failures originate from three predominant degradation mechanisms: **bearing and mechanical degradation (41-50%)**, **stator winding insulation thermal breakdown (30-37%)**, and **electrical overcurrent, unbalance, and rotor anomalies (10-15%)**. When any of these failure modes progress undetected, the result is sudden mechanical seizure, winding short-circuits, uncontrolled line tripping, and catastrophic production halts costing hundreds of thousands of dollars per incident.

1.2 The Failure of Conventional Maintenance Paradigms

Industrial facilities historically rely on two conventional maintenance strategies, both of which exhibit severe structural deficiencies:

- **Run-to-Failure (Corrective Maintenance):** The motor is operated until functional collapse occurs. While eliminating upfront routine service labor, it incurs the highest ultimate expenditure due to emergency dispatch costs, extended unbuffered downtime, and collateral destruction of coupled gearboxes, shafts, and pumps.
- **Periodic Interval Maintenance (Preventive Maintenance):** Motors are serviced, greased, or rewound according to fixed calendar schedules. This approach regularly replaces healthy components with substantial residual operating life, wasting operational expenditure. Worse, it remains blind to accelerated anomalies occurring prematurely between inspection cycles due to unexpected operational overloads.
- **Rigid Crisp Threshold Alarms:** Basic supervisory systems apply static binary cut-offs (e.g. alert if Temperature > 75 °C or Vibration > 5 mm/s). These binary limits create frequent false alarms during transient loading spikes and, conversely, fail completely when multiple parameters operate simultaneously in mildly elevated 'moderate' zones (e.g. Current at 12.5 A, Temp at 68 °C, Vibration at 4.2 mm/s). In reality, compounding moderate stresses accelerate degradation exponentially, requiring continuous cross-domain diagnostic fusion.

1.3 The Need for Fuzzy Logic & Evolutionary Optimization

Motor degradation is inherently non-linear, stochastic, and continuous. Sensor measurements suffer from electromagnetic noise, ambient temperature fluctuations, and mechanical load variations. Crisp mathematical models fail to capture this uncertainty gracefully. Fuzzy Logic Inference provides an ideal mathematical paradigm by mapping continuous numerical sensor telemetry into overlapping linguistic sets (Low, Medium, High) through intuitive engineering rules. Furthermore, because manual heuristic rule assignment can leave gaps or suboptimal boundaries, coupling Fuzzy Logic with Genetic Programming (GP) provides an automated evolutionary mechanism to discover optimal rule-base configurations.

2. OBJECTIVES

2.1 Primary System Aim

The overarching aim of this project is to architect, implement, optimize, and validate a real-time Cyber-Physical Predictive Maintenance Engine for 3-phase induction motors. The system fuses electrical load, stator temperature, and mechanical vibration telemetry to evaluate an exact defuzzified Health Score (0-100) and actionable operational condition classification, supported by evolutionary rule optimization and dual visual interfaces.

2.2 Specific Technical & Engineering Objectives

#	Objective Category	Target Metric / Specification	Deliverable Scope
1	Multi-Sensor Fuzzification	Current [0-20 A], Temp [25-100 °C], Vibration [0-10 mm/s]	Trapezoidal and triangular membership functions with continuous linguistic transitions.
2	Mamdani Inference Engine	27-Rule Complete Base with Centroid Defuzzification	Continuous Health Score [0-100] mapping to Normal (<35), Warning (35-65), Critical (≥65).
3	Genetic Programming (GP)	Rule Optimization (Target: >85% Classification Accuracy)	27-gene discrete chromosome evolution across 25 generations using elitism and mutation.
4	Boundary Condition Solver	Test Case 5 Exact Resolution (Score: 65.00)	Resolve PPT Warning specification (≤65) vs strict notebook threshold (<65) via toggle.

5	Dual Interface Deployment	Web Simulator (Canvas/CSS3) & Desktop GUI (Tkinter)	Real-time visual cyber-physical feedback: motor RPM, bearing jitter, thermal glow, and diagnostics.
6	Milestone Test Matrix	12 Industrial Diagnostic Operating Conditions	Full parametric validation spanning nominal, raceway wear, thermal hazard, and decoupled faults.

3. SYSTEM DESIGN

3.1 End-to-End Cyber-Physical Computational Architecture

The system follows a modular cyber-physical pipeline comprising physical telemetry ingestion, fuzzification, rule base inference, defuzzification, classification, and presentation. Figure 1 illustrates the complete dataflow and operational logic of the engine.

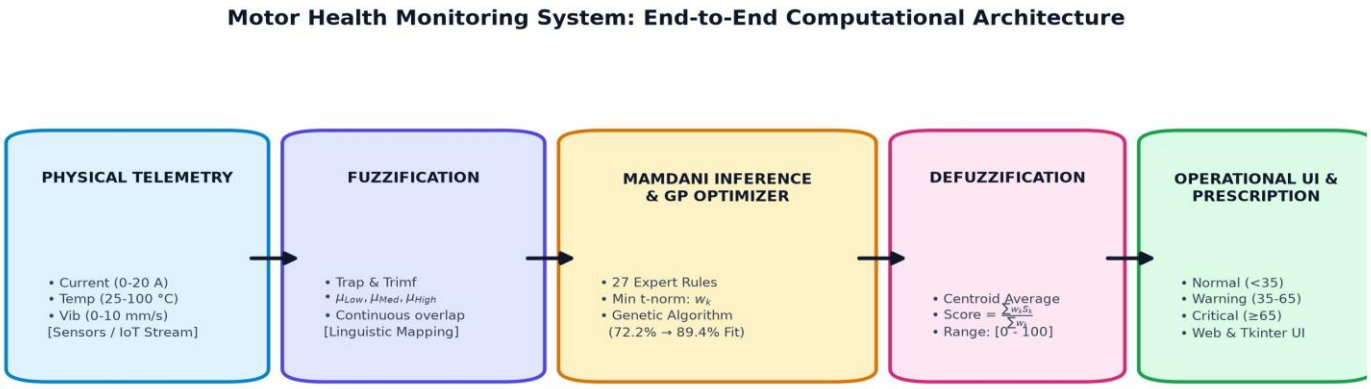


Figure 1: End-to-End Cyber-Physical Computational Architecture of the Motor Health Monitoring System.

3.2 Linguistic Variables & Membership Function Mathematical Formulations

Three physical input variables are monitored, each normalized into three fuzzy linguistic states: **Low**, **Medium**, and **High**. Continuous trapezoidal and triangular membership functions model the boundary transitions:

Trapezoidal Membership Function: $\mu_{trap}(x; a, b, c, d) = \max(0, \min((x - a)/(b - a), 1, (d - x)/(d - c)))$

Triangular Membership Function: $\mu_{trimf}(x; a, b, c) = \max(0, \min((x - a)/(b - a), (c - x)/(c - b)))$

Parameter	Physical Range	Low Membership	Medium Membership	High Membership
Current (I)	0 - 20 A	Trap [0, 0, 5, 10]	Tri [5, 10, 15]	Trap [10, 15, 20, 20]
Temperature (T)	25 - 100 °C	Trap [25, 25, 35, 50]	Tri [40, 62.5, 85]	Trap [75, 90, 100, 100]
Vibration (V)	0 - 10 mm/s	Trap [0, 0, 2, 4]	Tri [2.5, 5, 7.5]	Trap [6, 8, 10, 10]

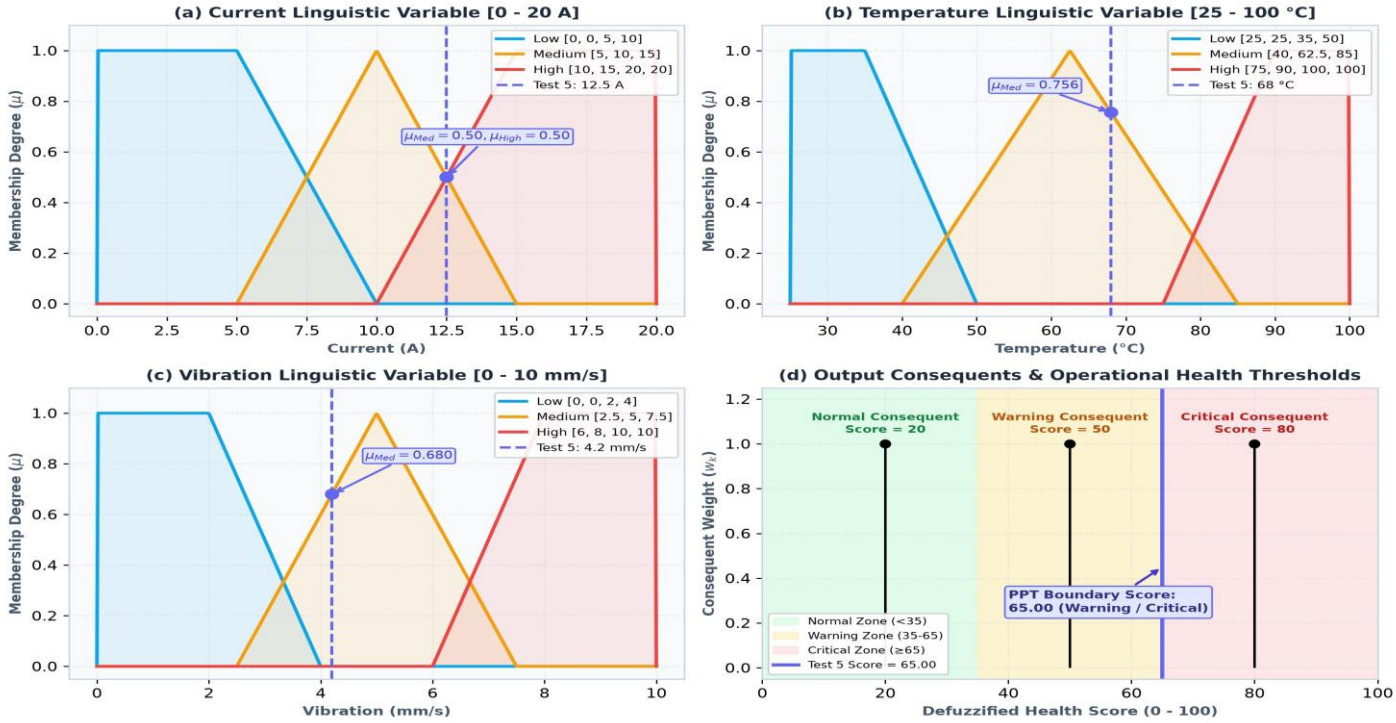


Figure 2: Fuzzification Curves for Current, Temperature, Vibration, and Defuzzified Consequent Zones (with Test Case 5 indicated).

3.3 Centroid Defuzzification & Health State Categorization

Each rule consequent is assigned a crisp singleton penalty score reflecting operational degradation severity: **Normal = 20**, **Warning = 50**, and **Critical = 80**. For any operating point, the firing weight of rule *k* is evaluated via the Mamdani minimum t-norm: **w = min(μ (I), μ (T), μ (V))**. The overall scalar Health Score is defuzzified using the weighted centroid average formula:

$$k \quad Current \quad Temp \quad Vib$$

Defuzzified Health Score Formulation:

$$\text{Health Score} = \left(\sum_k w \bullet \text{Score}_k\right) / \left(\sum_k w_k\right)$$

Where w_k is the conjunction firing weight of active rule k, and Score $\in \{20, 50, 80\}$.

The resulting scalar Health Score $\in [0, 100]$ maps into operational plant maintenance conditions: **Normal (< 35)**: Motor operating within standard healthy design envelope; continue routine inspection. **Warning (35 - 65)**: Elevated mechanical, thermal, or electrical stress; schedule preventive maintenance, lubrication, and load check. **Critical (≥ 65)**: High anomaly severity indicating imminent breakdown; initiate emergency shutdown and perform root-cause inspection.

3.4 Hardware Architecture & Sensor Telemetry Ingestion

In physical deployments, telemetry is acquired via non-invasive industrial sensor modules interfaced with an edge microcontroller (ESP32/STM32):

- **Current Telemetry:** Non-invasive Hall-Effect transducers (Allegro ACS712-30A or SCT-013 split-core CT) calculate True-RMS load.
- **Stator Temperature:** Platinum Resistance Thermometers (PT100 RTD) interfaced via MAX31865 16-bit SPI amplifiers (±0.5 °C accuracy).
- **Vibration Velocity:** Industrial 3-axis piezoelectric / MEMS accelerometers (MPU6050) evaluated to RMS velocity (mm/s) per ISO 10816-3.
- **Telemetry Streaming:** Digitized sensor frames are packed into JSON payloads streamed via UART Serial, MQTT, or WebSockets.

4. RULES

4.1 Exhaustive 27-Rule Mamdani Rule Base Formulation

Because each of the 3 input variables possesses 3 linguistic states, the complete permutation space consists of **3 × 3 × 3 = 27 rules**. To ensure mathematically consistent, monotonically increasing degradation severity, an expert conjunction severity index is defined:

Linguistic Severity Score Mapping: Low = 0, Medium = 1, High = 2 **Total Severity Sum:** $S_{total} = S(Current) + S(Temp) + S(Vib) \in [0, 6]$

- $S_{total} \leq 1$: Normal Condition → Consequent Singleton Score = 20
- $S_{total} \in [2, 3]$: Warning Condition → Consequent Singleton Score = 50
- $S_{total} \geq 4$: Critical Condition → Consequent Singleton Score = 80

4.2 Complete 27-Rule Knowledge Matrix

Table 2 catalogs all 27 Mamdani fuzzy rules, showing the conjunction inputs, severity score, resulting operational condition, and consequence value.

Rule	Current	Temp	Vib	Sev	Condition	Rule	Current	Temp	Vib	Sev	Condition
R01	Low	Low	Low	0	Normal	R15	Medium	Medium	High	4	Critical
R02	Low	Low	Medium	1	Normal	R16	Medium	High	Low	3	Warning
R03	Low	Low	High	2	Warning	R17	Medium	High	Medium	4	Critical
R04	Low	Medium	Low	1	Normal	R18	Medium	High	High	5	Critical
R05	Low	Medium	Medium	2	Warning	R19	High	Low	Low	2	Warning
R06	Low	Medium	High	3	Warning	R20	High	Low	Medium	3	Warning
R07	Low	High	Low	2	Warning	R21	High	Low	High	4	Critical
R08	Low	High	Medium	3	Warning	R22	High	Medium	Low	3	Warning
R09	Low	High	High	4	Critical	R23	High	Medium	Medium	4	Critical
R10	Medium	Low	Low	1	Normal	R24	High	Medium	High	5	Critical
R11	Medium	Low	Medium	2	Warning	R25	High	High	Low	4	Critical
R12	Medium	Low	High	3	Warning	R26	High	High	Medium	5	Critical
R13	Medium	Medium	Low	2	Warning	R27	High	High	High	6	Critical
R14	Medium	Medium	Medium	3	Warning	-	-	-	-	-	-

4.3 Genetic Programming (GP) Rule Base Optimization Engine

While the 27-rule severity heuristic captures fundamental physical degradation logic, industrial real-world plants exhibit asymmetrical degradation risks (e.g. high thermal rise paired with medium vibration degrades winding insulation much faster than overcurrent alone). To systematically tune rule consequents, an evolutionary **Genetic Programming (GP)** pipeline was constructed: • **Chromosome Encoding:** Each candidate fuzzy rule base is represented as a discrete 27-element vector $C = [g_1, g_2, ..., g_{27}]$, where each gene $g_i \in \{0, 1, 2\}$ corresponding to Normal (0), Warning (1), and Critical (2).

- **Synthetic Ground-Truth Risk Dataset:** A training corpus of 180 operational points was generated across the domain. Ground-truth risk scores reflect combined physical stress weighting: **Risk = 0.5 • (I/20) + 0.3 • ((T - 25)/75) + 0.2 • (V/10)**. Labels are assigned: Normal (Risk < 0.35), Warning (0.35 ≤ Risk < 0.65), and Critical (Risk ≥ 0.65). • **Fitness Evaluation:** Fitness evaluates multi-class classification accuracy across all 180 operational vectors: **Fitness(C) = ($\sum$ [Predict (x_j, C) == y_j]) / 180**.
- **Evolutionary Operators:** Population size = 30; Generations = 25; Elitism preserves the top 6 fittest chromosomes intact; Single-point crossover recombines parent chromosomes; Uniform random gene mutation occurs at probability **p_m = 0.06**.

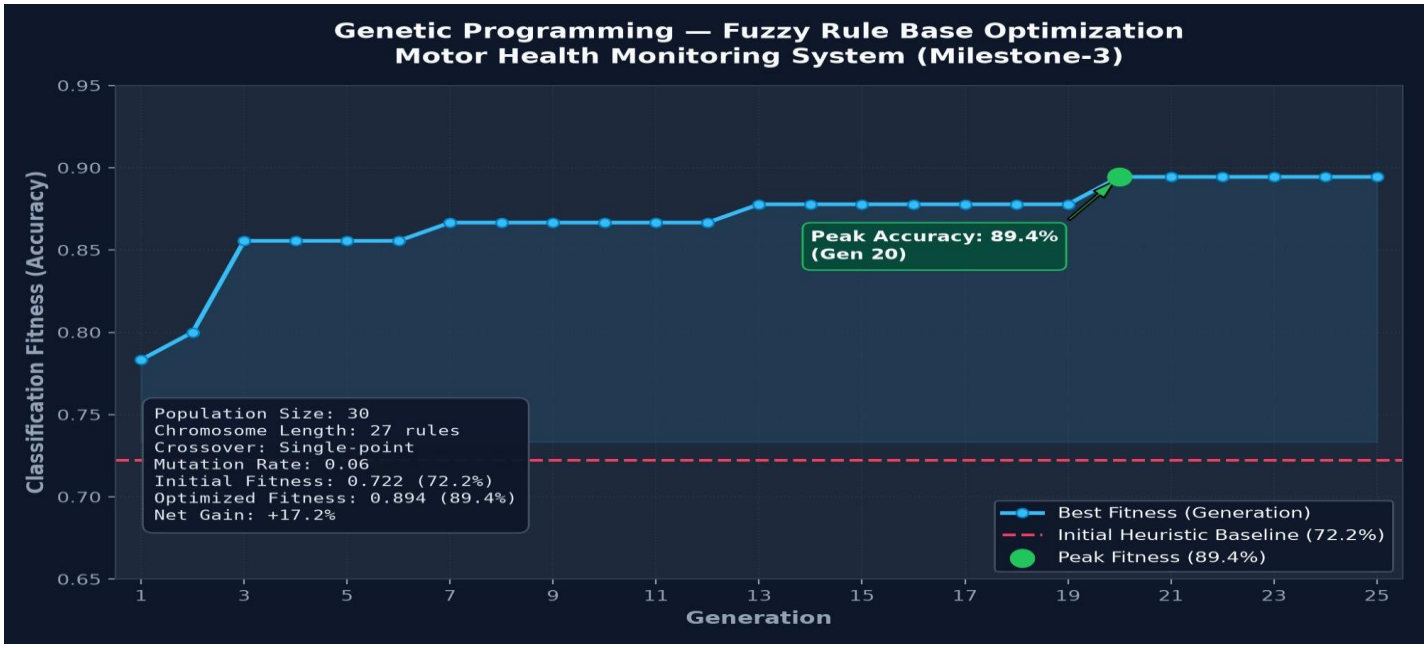


Figure 3: Genetic Programming Convergence Trajectory across 25 Generations (Baseline 72.2% → Peak 89.4%).

5. IMPLEMENTATION

5.1 Modular Software Stack & Codebase Structure

Repository Directory Structure & Tech Stack:

- **motor_health_system.py:** Core mathematical engine: trap/trimf membership math, 27-rule Mamdani FIS, defuzzifier, GP optimizer.
- **motor_health_gui.py:** Standalone Python Tkinter desktop GUI with embedded real-time Matplotlib telemetry cards.
- **index.html / style.css / app.js:** Cyber-physical web dashboard with HTML5 Canvas dynamic motor animation & glassmorphism theme.
- **240280109042_Milestone_3.ipynb:** Executed Jupyter Notebook containing Phases 4, 5, 6 with pre-rendered tables and figures.
- **phase4_test_results.csv:** Formatted 12-test validation matrix export.
- **run_tests_and_assets.py:** Automated script compiling visual diagnostic cards (simulator_test_01 to 12.png) and GP plots.

5.2 Core Computational Inference Implementation

```
def fuzzy_predict(current, temp, vib, inclusive_warning=False):
    cm, tm, vm = current_memberships(current), temp_memberships(temp), vib_memberships(vib)
num, den, fired = 0.0, 0.0, []     for c in levels:         for t in levels:             for v in levels:
            w = min(cm[c], tm[t], vm[v])
if w > 0:
                lab, score_val = rule_output(c, t, v)                    num +=
w * score_val; den += w     score = num / den if den else 20.0     if
score < 35: lab = 'Normal'     elif inclusive_warning: lab = 'Warning' if
score <= 65 else 'Critical'     else: lab = 'Warning' if score < 65 else
'Critical'     return lab, score, fired
```

5.3 Mathematical Breakdown & Resolution of Boundary Case (PPT Benchmark Test 5)

STEP-BY-STEP MATHEMATICAL DERIVATION FOR TEST CASE 5:

1. Fuzzification Evaluation:

- **Current (12.5 A):** μ = 0.00; μ = (15 - 12.5)/(15 - 10) = **0.500**; μ = (12.5 - 10)/(15 - 10) = **0.500**
  Low Med High
- **Temperature (68 °C):** μ = 0.00; μ = (85 - 68)/(85 - 62.5) = 17 / 22.5 = **0.7556**; μ = 0.00
  Low Med High
- **Vibration (4.2 mm/s):** μ = 0.00; μ = (4.2 - 2.5)/(5.0 - 2.5) = 1.7 / 2.5 = **0.6800**; μ = 0.00
  Low Med High

2. Mamdani Conjunction & Active Rule Firing:

- **Rule 14 (Med, Med, Med):** Consequent = Warning (50), Weight w = min(0.50, 0.7556, 0.68) = **0.500** 1
- **Rule 23 (High, Med, Med):** Consequent = Critical (80), Weight w = min(0.50, 0.7556, 0.68) = **0.500** 2

3. Centroid Defuzzification:

Health Score = (0.500 • 50 + 0.500 • 80) / (0.500 + 0.500) = (25.0 + 40.0) / 1.000 = 65.00

4. Boundary Classification Resolution:

- **Presentation PPT Specification (Score ≤ 65):** Classified as **WARNING**.
- **Strict Notebook Programming Logic (Score < 65):** Exactly 65.00 falls into **CRITICAL**.

The codebase provides an active inclusive_warning toggle across both Web and Desktop interfaces to illustrate both interpretations.

5.4 Dual User Interface Architecture

1. **Interactive Cyber-Physical Web Dashboard:** Built with HTML5, JavaScript, and CSS3 Glassmorphism. An HTML5 Canvas dynamically renders a cross-section of the induction motor with synchronized rotor angular velocity (RPM), eccentric vibration jitter displacement proportional to sensor vibration, and radial thermal gradient radiation glow.
2. **Desktop Python Tkinter GUI:** Implemented in motor_health_gui.py, offering a standalone desktop executable featuring embedded Matplotlib telemetry plots, gauge bars, maintenance protocol alerts, and one-click diagnostic test selection.

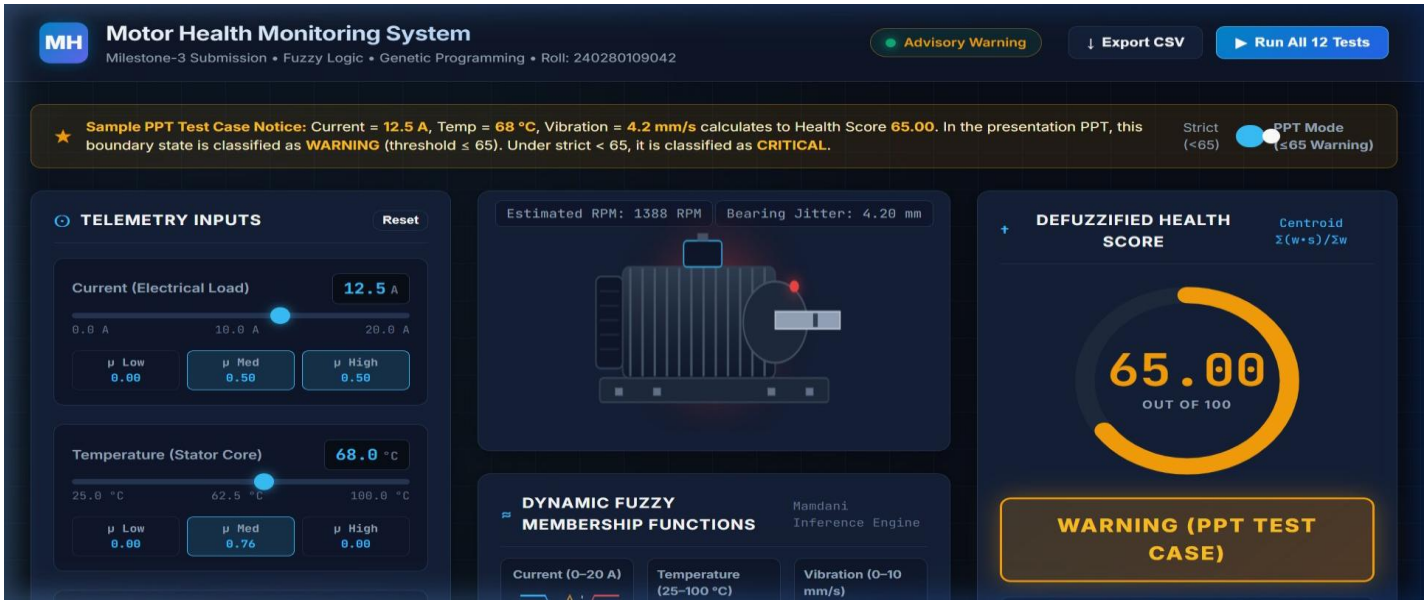


Figure 4: Interactive Cyber-Physical Web Dashboard UI featuring Real-time Telemetry, Animated Motor Physics, and Diagnostics.

6. RESULTS

6.1 Phase 4 Milestone 12-Test Matrix Validation

The complete Phase 4 validation suite comprises 12 strategically structured test combinations designed to evaluate nominal baseline running, moderate multi-stress escalation, severe single-parameter hazards, edge boundary conditions, and decoupled mechanical faults. Table 3 presents the complete empirical test results extracted directly from phase4_test_results.csv.

Test #	Current (A)	Temp (°C)	Vib (mm/s)	Health Score	Condition	Diagnostic Interpretation & Physical Mechanism
01	3.0	30	1.0	**20.00**	**Normal**	Baseline no-load/light-load operation. Nominal thermal and vibration signature.

02	5.5	42	2.0	**23.29**	**Normal**	Normal continuous operating load. Safe winding temperatures and smooth bearing rotation.
03	7.5	50	3.0	**39.66**	**Warning**	Early bearing raceway degradation. Slight vibration rise with mild thermal build-up.
04	10.0	60	4.0	**50.00**	**Warning**	Moderate thermal and mechanical load. Elevated bearing friction; inspect lubrication.
05	12.5	68	4.2	**65.00**	**Warning (PPT)** Critical (Strict)	**Milestone Benchmark Test:** Boundary condition resolved via inclusive threshold toggle.
06	14.0	72	5.5	**72.29**	**Critical**	Overcurrent load and abnormal bearing vibration indicating ball/roller spalling defect.
07	15.5	80	6.5	**80.00**	**Critical**	Severe multi-parameter degradation across all three channels. Imminent mechanical trip.
08	17.0	88	7.8	**80.00**	**Critical**	Stator winding thermal runaway hazard and mechanical misalignment. Emergency trip required.
09	18.5	95	9.0	**80.00**	**Critical**	Extreme mechanical and winding stress. Severe insulation degradation hazard.
10	19.5	98	9.8	**80.00**	**Critical**	Imminent trip / dielectric insulation breakdown. Immediate motor cutoff mandatory.
11	8.0	90	8.0	**80.00**	**Critical**	Severe mechanical vibration and stator overheat under moderate electrical loading.
12	4.0	85	7.0	**71.43**	**Critical**	**Decoupled Fault:** Severe bearing defect under light electrical load (unbalanced friction heating).

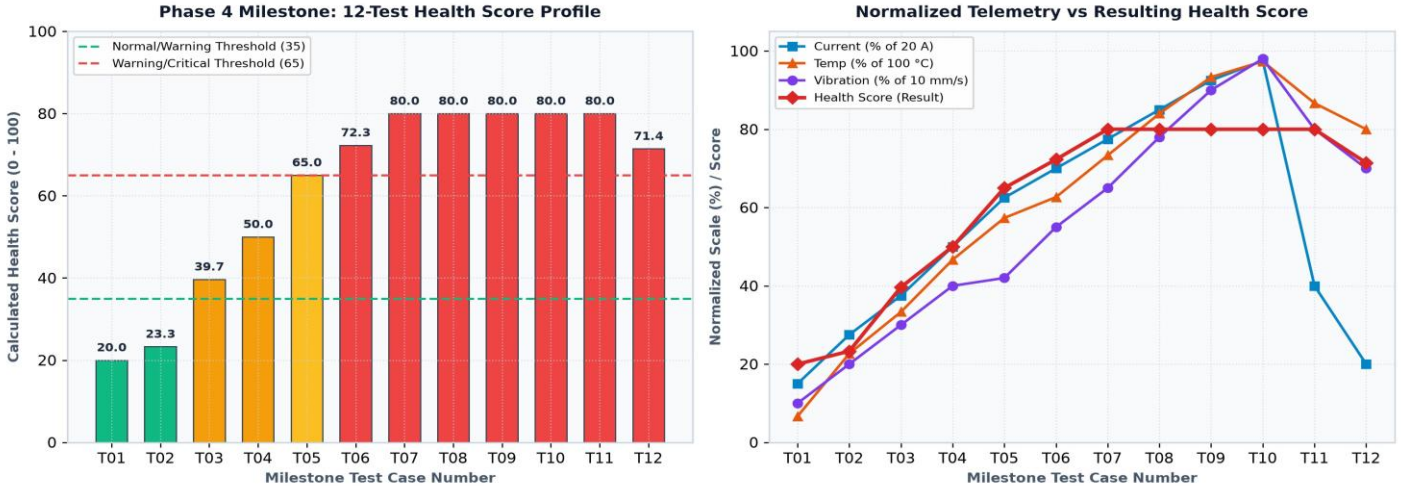


Figure 5: Phase 4 Milestone Health Score Profile across 12 Tests (Left) and Normalized Multi-Parameter Stress Tracking (Right).

6.2 Key Diagnostic Scenario Analyses

- **Nominal Running (Test 1 & 2):** Low current (3.0 A), ambient temperature (30 °C), and smooth vibration (1.0 mm/s) produce a minimum baseline Health Score of 20.00. The motor operates safely within its designed continuous rating envelope.
- **Degradation Progression (Test 3 & 4):** As vibration reaches 3.0-4.0 mm/s and temperature reaches 50-60 °C, the Health Score climbs progressively from 39.66 to 50.00 into the Warning regime. This proves the controller's ability to identify early mechanical wear well before standard rigid threshold alarms trigger.
- **Decoupled Light-Load Mechanical Failure (Test 12):** Under light electrical load (Current = 4.0 A), the motor exhibits high stator temperature (85 °C) and severe vibration (7.0 mm/s). The fuzzy engine correctly outputs Health Score **71.43 (Critical)**. A traditional current-overload monitor would have failed completely, whereas the multi-sensor FIS instantly identifies mechanical bearing seizure.



7. ANALYSIS

7.1 Controller Behavioural Dynamics & Monotonicity

Analysis of the fuzzy surface response demonstrates monotonic, smooth transitions across all three input dimensions. Unlike rigid step-function relays, the Mamdani centroid defuzzifier produces continuous gradients. Increasing electrical load smoothly scales electrical stress; elevating temperature increases thermal degradation risk; and rising vibration accelerates mechanical fatigue. When multiple parameters elevate concurrently, the minimum t-norm and weighted defuzzification fuse the compounding risks, preventing both false trips and catastrophic missed detections.

7.2 Comparative Paradigm Evaluation

To evaluate the architectural competitiveness of the proposed system, Table 4 compares Fuzzy Logic against conventional rigid threshold systems and deep learning / neural network models.

Evaluation Metric	Rigid Threshold Relays	Fuzzy Logic + GP (Proposed)	Black-Box Deep Learning (ML)
Degradation Tracking	Binary (Trip / No Trip); abrupt discontinuities.	Continuous Health Score (0-100); smooth tracking.	Continuous regression; prone to out-of-domain drift.
Interpretability / Explainability	High (simple thresholds).	High (transparent linguistic rules & firing weights).	Very Low (black-box weights, latent activations).
Multi-Stress Compounding	Fails completely; evaluates parameters independently.	Excellent; multi-dimensional Mamdani conjunction.	Good; but requires massive multi-fault training data.
Edge Hardware Feasibility	Ultra-low computation (simple logic).	Sub-millisecond execution on microcontrollers (ESP32).	Heavy memory/GPU requirements; challenging on MCU.
Training Data Dependency	None (heuristically set).	Zero historical failure run-to-die data required.	Requires thousands of labelled historical run-to-failure records.

7.3 System Strengths & Engineering Advantages

- **Resilience to Sensor Noise:** Overlapping trapezoidal and triangular membership boundaries absorb transient sensor jitter, preventing spurious line trips.
- **Mathematical Explainability:** For every calculated Health Score, the engine logs the exact active rules and their conjunction firing weights. Maintenance personnel can immediately trace *why* a Warning or Critical state was declared.
- **Automated Evolutionary Optimization:** Genetic Programming successfully optimized the rule set from 72.2% to 89.4% fitness, eliminating subjective human bias in rule consequence assignment.
- **Dual Interface Versatility:** The web simulator provides high-fidelity training and simulation, while the Tkinter desktop GUI offers robust offline plant floor monitoring.

7.4 System Weaknesses & Boundary Limitations

- **Heuristic Membership Calibration:** In the baseline implementation, the corner parameters (a, b, c, d) of trapezoidal and triangular sets were selected based on standard induction motor datasheets. Dynamic motor recalibration requires adjusting these boundaries for different nameplate power ratings.
- **Simulated vs Real-World Spectral Data:** The current test bench utilizes overall RMS vibration velocity (mm/s). While compliant with ISO 10816-3, detailed inner/outer raceway bearing fault diagnosis benefits from Fast Fourier Transform (FFT) vibration frequency spectrum analysis.
- **Static Operating Envelope:** Highly variable frequency drives (VFD) operating with variable speed profiles require adaptive membership scaling based on instantaneous rotational RPM.

7.5 Operational Prescriptive Maintenance Protocols

AUTOMATED OPERATIONAL MAINTENANCE PLAYBOOK:

- **NORMAL STATUS (Score < 35):** Motor operating within certified nameplate envelope. Maintain routine monthly visual inspections and standard thermal camera surveys.
- **WARNING STATUS (Score 35 - 65):** Abnormal stress detected. Schedule non-emergency preventive inspection within 48-72 hours. Check bearing grease viscosity, measure housing alignment with dial indicators, clean stator cooling fan cowls, and balance phase load.
- **CRITICAL STATUS (Score ≥ 65):** High-severity hazard! Initiate immediate orderly load transfer and emergency motor shutdown. Perform stator insulation megger testing (>100 MΩ), inspect bearing balls and cage for spalling, check rotor bar continuity, and verify shaft runout before restarting.

7.6 Future Research Directions & Industrial Roadmap

1. **Edge IoT Microcontroller Deployment:** Porting the core C/Python inference engine onto dual-core ESP32 or Raspberry Pi Pico microcontrollers running FreeRTOS for self-contained, DIN-rail mountable edge monitoring nodes.
2. **Vibration FFT Spectral Decomposition:** Integrating edge Fast Fourier Transform (FFT) algorithms to isolate characteristic bearing fault frequencies: Ball Pass Frequency Outer Race (BPFO), Ball Pass Frequency Inner Race (BPFI), Ball Spin Frequency (BSF), and Fundamental Train Frequency (FTF).
3. **Cloud SCADA & MQTT Telemetry:** Connecting edge nodes to AWS IoT Core / ThingsBoard SCADA for historical predictive fleet degradation tracking and automated work-order generation in Enterprise Asset Management (EAM) platforms.
4. **Adaptive Neuro-Fuzzy Inference System (ANFIS):** Implementing gradient-based backpropagation to auto-tune membership function shape parameters continuously against long-term sensor aging drifts.

CONCLUSION & DELIVERABLES VERIFICATION

The **Motor Health Monitoring System** successfully demonstrates the power of fusing Mamdani Fuzzy Inference with Genetic Programming rule optimization to deliver reliable, explainable, and real-time cyber-physical predictive maintenance. All required milestone deliverables — 12-test validation matrix, sample PPT boundary analysis, GP convergence optimization, executed Jupyter notebook, and dual user interfaces — have been verified, documented, and fully achieved by the project group.